

Record Review: Zenodo 22308897

Calculated.Motion.0001.mp4 — DOI 10.5281/zenodo.22308897 — community: Rock Street

Review date: 4 September 2026. Reviewer: Claude (Anthropic), at the request of Attogram / Provo 42. Purpose: this record is a pilot for a larger DOI series; the review lists everything verifiable about the record and file, then critiques it so the next deposits can be templated correctly.

Method. Zenodo landing page fetched and read; the deposited file was supplied separately and its MD5 verified against the record (match). Container probed with ffprobe; subtitle track extracted with ffmpeg; frames sampled every 3 s plus first/last frames and pixel-diffed. No claims below rest on memory of the record — everything is from the page or the bytes.

Part 1 — What the record says (Zenodo metadata)

Title	Calculated.Motion.0001.mp4
DOI	10.5281/zenodo.22308897
Published	4 September 2026, version v1 (only version)
Creator	“Attogram, Provo 42” — entered as a single person name (family: Attogram, given: Provo 42)
Description	“Calculated.Motion.0001.mp4” (the filename, verbatim)
Resource type	Video/Audio
Access	Open
Community	Rock Street
Licenses	Creative Commons Attribution 4.0 International and MIT License (two entries, no stated relationship)
Keywords / subjects	None
Related identifiers	None (no isPartOf, isDerivedFrom, isVersionOf, etc.)
Version notes	None
Language	Not set
Funding / grants	None
Files	1 file, Calculated.Motion.0001.mp4, 3.4 MB, md5 b84c2600faf8234bf61f219c0b38d14e
Stats at review	0 views, 0 downloads
Indexed in	OpenAIRE

Part 2 — What the file actually contains (technical inspection)

Container and streams

Local filename	Calculated_Motion_0001.mp4 (underscores; Zenodo copy uses dots) — bytes identical, MD5 match
Container	ISO MP4 (isom / iso2 / avc1 / mp41), muxed by Lavf59.27.100 (FFmpeg)
Duration	30.80 s
Video stream	H.264 via libopenh264 (Lavc59.37.100), 1024 × 1024 px, 30 fps, 924 frames, ~681 kb/s
Audio stream	AAC, ~198 kb/s, 30.77 s
Subtitle stream	mov_text, 11 cues, 0–30 s (full lyric, see below)
Container tags	title = “Calculated Motion”; album = “Geometric Hearts”; genre = “Progressive J-Pop”
Overall bitrate	~889 kb/s

Visual content

The picture is a **single static frame** held for roughly the first 27 seconds: generated cover art showing a bass guitar over blueprint-style schematics and formula fragments, with a “Calculated Motion” wordmark in the lower

right. A pixel difference between the frame at $t=0$ and $t=21$ s returns an empty bounding box — there is no animation, zoom, or pan in the artwork. The final ~4 seconds are a **Google Gemini outro**: a black card with the animated Gemini sparkle logo and the text “gemini.google.com/music”.

Sampled frames at 3 s intervals (0–27 s), then first frame, mid, late, and the outro:



Figure: contact sheets extracted from the deposited file. Frames 1–9 are pixel-identical; frame 10 and the bottom-right two panels are the Gemini outro.

Embedded subtitle track (lyric)

Extracted verbatim from the mov_text stream. This text exists nowhere in the Zenodo metadata.

```
00:00 Three months of data, a meticulous blueprint.
00:03 Every parameter locked in the sequence.
00:06 This velocity has a structural reason,
00:09 a logic beyond the rhythm of the season.
00:12 You might hear the chaos of sound and light,
00:15 but it's the iterative process taking flight.
00:18 It's a beautiful contradiction I embrace,
00:21 in this hyper-calculated digital space.
00:24 THE MUSIC MAY BE RANDOM, BUT THE METHOD IS NOT.
00:27 (THE METHOD IS NOT.)
```

Provenance inferred from the bytes

- The outro card, the 1024×1024 square format, the 30 s length and the container tag set are consistent with a direct export from Google Gemini's music feature. Nothing in the record states this.

- The cover artwork is almost certainly generated in the same pipeline (it carries the track title as a wordmark). No prompt, model, date, or seed is recorded anywhere.
- The file was re-muxed through FFmpeg (Lavf) with libopenh264 — i.e. it has passed through at least one local tool after generation. That step is also undocumented.

Part 3 — Critique, ordered by severity

3.1 Third-party branding is inside a CC-BY-licensed file (blocking)

The last ~4 s of the file are Google's Gemini logo and URL. The record licenses the whole file under CC BY 4.0 (and MIT). Neither license can grant rights over Google's trademarks; a downstream reuser who trusts the license inherits a problem the record created. This is the same class of exposure as the Big Bird incident, at lower stakes. Fix: trim the outro before deposit (new bytes = new version), or at minimum state in the description that the outro is third-party branding excluded from the license grant. Trimming is cleaner.

3.2 Ownership of generated output is unverified (blocking for the series, not this record)

Before CC-licensing any Gemini-generated audio or artwork, confirm what Google's current terms say about ownership and permitted redistribution of generated content, and whether an audible/visual attribution requirement exists. This is the real legal question underneath the license-selection question. Check it once, write the finding into the corpus, and template the license line from it. (Reviewer is not a lawyer.)

3.3 Dual license with no operator (high)

Two licenses are attached with no stated relationship. Zenodo's rights field, and the DataCite rightsList it exports to, are flat lists — there is no AND/OR. A reader cannot tell whether this is “reuser's choice” or “both apply”. MIT is also a software license whose text speaks of “the Software” and source code; on a media file it is noise. It also contradicts the project's own tri-license policy (MIT for code, CC BY for creative artifacts). Fix: CC BY 4.0 only for media. If dual licensing is truly intended, put the SPDX expression MIT OR CC-BY-4.0 as the first line of the description and ship a LICENSE file inside the record.

3.4 No AI-provenance statement (high)

Nothing on the record says the audio, lyric, or artwork are machine-generated, by which tool, or when. For a project whose thesis is receipts, this is the most on-brand omission. Fix: a standard provenance block (tool, version/date, human edits, post-processing) in every generated-media record.

3.5 Description is the filename (high)

The abstract field contains only “Calculated.Motion.0001.mp4”. OpenAIRE, Google Dataset Search, and community search get nothing to index. The irony: the file already carries a title, album, genre and the full lyric in its container and subtitle stream — better metadata than the record. Fix: lift them.

3.6 No relationship edges (medium)

The “0001” suffix and the “Geometric Hearts” album tag both imply a sequence and a parent, but the record has zero related identifiers. The project's own edge taxonomy (isPartOf, isDerivedFrom, isDocumentedBy, etc.) is unused. Fix: isPartOf → a collection/album record; isDocumentedBy → the GitHub issue; isDerivedFrom → source audio if a canonical audio-only artifact exists.

3.7 Creator entered as a person named “Provo 42 Attogram” (medium)

Zenodo parsed the creator as family name “Attogram”, given name “Provo 42”. Citations will render accordingly. Fix: one person creator (Attogram, with ORCID if available) plus one organisation creator or affiliation (Provo 42 / the Stichting once registered). Decide this once — it becomes the citation string for every DOI in the series.

3.8 Resource type and canonical form (medium)

This is an audio work in a video wrapper: a still image plus a track. Depositing the mp4 as the only artifact makes the audio harder to reuse (no one wants a 3.4 MB mp4 to get a 30 s clip). Fix: consider an audio-only file (m4a/flac) as the canonical artifact, with the mp4 as a derivative, or deposit both in one record. Either way, say what the canonical form is.

3.9 Missing low-cost fields (low)

- Keywords: none. At minimum: Provo 42, Attogram, generated music, Geometric Hearts, Progressive J-Pop.
- Language: not set (lyric is English).
- Version notes: none. Even “v1 — raw export, outro not trimmed” is a receipt.
- Dates: only the publication date. Creation/generation date is absent.
- Filename convention drift: dots on Zenodo, underscores locally. Pick one for the corpus.

Part 4 — Proposed template for the next records

Suggested v2 description for this record

Calculated Motion (Geometric Hearts, track 0001)

AI-generated audio track with generated static cover art. Generated with Google Gemini (music feature), September 2026; re-muxed with FFmpeg; Gemini outro trimmed. Genre tag: Progressive J-Pop. Duration 30 s. Lyric (embedded as a subtitle track):

Three months of data, a meticulous blueprint. / Every parameter locked in the sequence. / This velocity has a structural reason, / a logic beyond the rhythm of the season. / You might hear the chaos of sound and light, / but it's the iterative process taking flight. / It's a beautiful contradiction I embrace, / in this hyper-calculated digital space. / The music may be random, but the method is not.

Part of the Provo 42 corpus (Attogram Project). Collection: Geometric Hearts [DOI]. Tracking issue: [GitHub URL].

License: CC BY 4.0 for the audio, lyric and artwork as published by Attogram. No claim is made over third-party marks or tools. Provenance and human edits are recorded above.

Metadata checklist per record (gate items for the RR pass)

Field	Rule
Title	Human title, not filename. Include series and index: “Calculated Motion (Geometric Hearts 0001)”
Creators	Person + organisation, fixed once for the whole series; ORCID if available
Description	What it is; how it was made (tool, date, human edits); lyric/transcript if any; where it sits in the corpus
Resource type	Audio for music with a still frame; Video only if there is actual motion
License	Exactly one per artifact type. Media = CC BY 4.0. Code = MIT. Never both on one media file
Provenance	Mandatory block for any generated content: tool, version/date, prompt (or pointer to it), post-processing
Third-party content	Scan every frame and the audio tail for logos, marks, likenesses before upload (RR gate)
Related identifiers	isPartOf (collection), isDocumentedBy (issue), isDerivedFrom (source) as applicable
Keywords	Project name, series name, genre, “generated media”, method keywords
Language	Set it
Version notes	One line per version, always
Files	Canonical file first; derivatives named as such; consistent filename convention; LICENSE.txt inside the record

Verdict

Valid DOI, valid file, valid community, and a lyric that is a compact statement of the project's method. As a pilot it succeeds at proving the pipeline works. As a record it is an unlabelled box: the file knows more about itself than the catalogue does, and it ships a third-party logo under a license that cannot cover it. Fix the branding and the provenance statement first; the rest is template work that pays off across every subsequent DOI.